Orbit Wars — Critic-Free RL Math, v12

The GRPO trainer: the four original tricks [A]–[D], the Tier 0–2 anti-collapse roadmap *(which empirically FAILED — legacy std-norm is the keeper)*,

and the pending Tier 3 structural changes

2026-06-19

EMPIRICAL VERDICT (2026-06-19 — read this first). The Tier 0–2 "anti-collapse" estimators specified in Part II (rank [C1], Dr. GRPO [B], sibling [B1], phi-value [C], cvar [C2], clip-higher [D3]) were tested in a controlled RTX 3070 Ti ablation and **ALL FAILED** — every one drove the agent into the *passivity* basin (launch-rate $\to 0$, Elo down), while plain **legacy per-group std-norm stayed stable and improving** (Section 14). The deterministic bound proof (Section 13) is correct but *bounded* $\neq$ *good*: rank discards win-*magnitude* and stops rewarding aggression. And the it426→it526 collapse this note opens with was a reward **TRIGGER** (an asymmetric "lose-slower" outcome decay), *not* the std-norm "amplifier" the early sections blame. **Keeper config: legacy std-norm + symmetric decay** (`WIN_DECAY` = `LOSS_DECAY` = 1, the v12 default). The theory in Sections 1–12 is preserved for the record and as a research menu — but it is contradicted by experiment; read Section 14 before trusting any of it.

This note specifies the four changes added to the v12 *critic-free* (GRPO) trainer, the math behind each, and the failure they fix. It is a companion to `rl_math_v9.tex` (the policy, reward, league, and PPO + GAE update are defined there and not repeated). Each trick is gated by a `Config` flag and **defaults to the legacy behaviour**, so a run is bit-for-bit unchanged until a flag is flipped; ablate them one at a time. The four:

tag	flag	idea
[A]	`GRPO_ANALYTIC_ADV`	closed-form 2p advantage at a shrunk win-rate (Section 2)
[B]	`GRPO_STD_NORM`, `GRPO_LOO`	drop the per-group std; leave-one-out baseline (Section 3)
[C]	`GRPO_PHI_VALUE`	use the shaping potential as a value, not a reward (Section 4)
[D]	`GRPO_CRN`	common random numbers across a world-group (Section 5)

Part II (Sections 6–15) — the Tier 0–2 anti-collapse roadmap. A *second* collapse (a warm-started GRPO run, elo2 2124 → 1772 → losing to its own snapshot *and* to the scripted `greedy`) prompted a broader set of levers, again all flag-gated and default-off. They attack the same amplifier from five established directions and pass a deterministic bound probe (Section 13). **But a controlled GPU ablation (Section 14) shows they do NOT work — every one of them collapses the agent into passivity, and the real it426 culprit was a reward TRIGGER, not the estimator. Read Section 14 before using any of them.** A tail of **pending Tier 3 structural** items follows (Section 15).

tier	tag	flag(s)	idea
0	**[D3]**	`CLIP_HI`	asymmetric upper clip (DAPO clip-higher), anti-passivity
0	**[D1]**	`RATIO_LENGTHNORM`	length-normalise the ratio (GSPO), planet-count-invariant clip/KL
0	**[D4]**	`ADV_TARGET_RMS`	pin the centred-advantage scale (no stealth LR drift)
1	**[C1]**	`GRPO_RANK_ADV`	within-group *rank* advantage: bounded $\sqrt{3}$, scale-free (Section 6)
1	**[A1]**	`GRPO_RB_GATE`	Rao–Blackwellise the launch gate (Section 8)
2	**[C2]**	`GRPO_CVAR_ALPHA`	risk-sensitive (CVaR) tail baseline (Section 9)
2	**[B1]**	`GRPO_SIBLING_BASELINE`	per-step credit from shared-root siblings (Section 10)
2	**[B2]**	`GRPO_AUX_VALUE`	train the dead value head as a representation aux (Section 11)
2	**[E1]**	`GRPO_OPP_BASELINE_W`	opponent-conditioned baseline — no-op until [E2] (Section 12)

Contents

1 The critic-free estimator and the collapse it produced

Variable reference — this section	
B	parallel envs per iteration, $B = n_W \cdot G$
G	*group size*: rollouts sharing one world (`GRPO_GROUP`); n_W = groups/iter
w_g	the world (initial planets/seating) shared by all G members of group g
R_i	rollout i's scalar return (full shaped return, or terminal-only if `GRPO_OUTCOME_ONLY`)
O_i	terminal outcome of rollout i; in 2p $O_i \in \{-1, 0, +1\}$, in 4p the placement-linear value
μ_g, σ_g	within-group sample mean / std of $\{R_i\}$
$\widehat{A}_i$	the advantage assigned to *every* timestep of rollout i
ε	`GRPO_ADV_EPS`, the std floor (10^{-4})
π_θ	the policy; a_t^i, s_t^i the action/state at step t of rollout i

GRPO replaces PPO's learned value baseline with a **group baseline**. Each iteration draws n_W worlds and rolls each out G times (the rollout lays the B envs out as contiguous blocks of G, one world per block, so a block is a group). With the return R_i standardized inside its group,

$$\widehat{A}_i = \frac{R_i - \mu_g}{\sigma_g + \varepsilon}, \qquad \mu_g = \tfrac{1}{G}\sum_{j \in g} R_j, \quad \sigma_g = \sqrt{\tfrac{1}{G}\sum_{j \in g}(R_j - \mu_g)^2}, \tag{1}$$

the gradient is the clipped surrogate of the PPO section applied with $\widehat{A}_i$ *flat over all timesteps* of rollout i: $\hat{g} = \mathbb{E}\big[\sum_t \nabla_\theta \log \pi_\theta(a_t^i \mid s_t^i)\, \widehat{A}_i\big]$. There is no critic, no GAE, and (with `GRPO_KL_COEF`$= 0$) no reference anchor.

The collapse. A warm-started run climbed to elo2 $\approx$ 2110, then over $\sim$40 iterations decayed to win-rate 0.09 with the launch rate sliding 0.80 $\rightarrow$ 0.30 — the project's familiar *passivity basin*. The trigger was an asymmetric outcome decay (a small "lose slower" term), but the *amplifier* was Eq. (1) itself: in a group that loses every rollout, σ_g collapses and the ε-floored division blows a trivial within-group tiebreak up to a full-magnitude "stall" gradient. Sections 2–3 dissect and remove that amplifier; Section 4 restores the per-step credit assignment GRPO structurally lacks; Section 5 tightens the baseline. All four are properties of *this* MDP: a terminal-dominated, near-binary reward evaluated on shared-world groups under self-play.

Empirical reversal (Section 14, added after the ablation). The framing above is half wrong. Removing the "amplifier" (Sections 2, 3, 6) turned out to be the *wrong* fix — it causes passivity, because the std-norm rare-win amplification is double-edged: it is also what keeps the agent *aggressive*. The right fix is to remove the *trigger* (keep the outcome decay symmetric); the identical legacy std-norm is stable without it. Treat Sections 2–12 as the mechanism/derivations behind a research menu, not as fixes to apply — the verdict is the red box on page 1 and Section 14.

2 [A] The 2p advantage in closed form, and its shrinkage

Variable reference — this section	
B_o	outcome magnitude in return units, B_o = `WIN_BONUS/PPO_REWARD_SCALE` (= 10)
w	number of *wins* among the G rollouts of a group
$\hat{p}$	empirical group win-rate w/G
$\tilde{p}$	Beta-shrunk win-rate, Eq. (4)
α	`GRPO_ANALYTIC_ALPHA`, the Beta pseudocount
$\widehat{A}_{\text{win}}, \widehat{A}_{\text{loss}}$	the two advantage values a 2p group can take

In 2p the outcome is binary, so (ignoring draws and any decay) every $R_i \in \{+B_o, -B_o\}$. A group with w wins, $\hat{p} = w/G$, has *exact* statistics

$$\mu_g = B_o(2\hat{p} - 1), \qquad \sigma_g = 2B_o\sqrt{\hat{p}(1-\hat{p})}, \tag{2}$$

and substituting into Eq. (1) (with $\varepsilon \to 0$) the standardized advantage takes **only two values**, a pure function of the win-rate:

$$\boxed{\widehat{A}_{\text{win}} = \sqrt{\frac{1-\hat{p}}{\hat{p}}}, \qquad \widehat{A}_{\text{loss}} = -\sqrt{\frac{\hat{p}}{1-\hat{p}}}} \tag{3}$$

Derivation. $\widehat{A}_{\text{win}} = \dfrac{B_o - \mu_g}{\sigma_g} = \dfrac{2B_o(1-\hat{p})}{2B_o\sqrt{\hat{p}(1-\hat{p})}} = \sqrt{\frac{1-\hat{p}}{\hat{p}}}$, and symmetrically for the loss. Two consequences:

- **Rare events are up-weighted.** As $\hat{p} \to 0$ (a world you almost always lose) a rare win gets $\widehat{A}_{\text{win}} \to +\infty$ while the common losses get $\widehat{A}_{\text{loss}} \to 0^-$: GRPO implicitly chases the rare win. This is a useful exploration prior — when it is real.
- **Every group carries unit gradient energy.** $\hat{p}\,\widehat{A}^2_{\text{win}} + (1-\hat{p})\,\widehat{A}^2_{\text{loss}} = (1-\hat{p}) + \hat{p} = 1$ for *all* $\hat{p}$. A 99% blowout pushes the gradient exactly as hard as a 50/50 toss-up.

The collapse lives at $\hat{p} \in \{0, 1\}$: then $\sigma_g = 0$, the division falls back on ε, and whatever infinitesimal residual distinguishes the (all-losing) rollouts — a length-dependent decay term, sampling noise — is renormalised to magnitude 1. The estimator manufactures a confident gradient out of noise.

The fix: evaluate Eq. (3) **at a shrunk $\tilde{p}$.** Replace the empirical $\hat{p}$ by a Beta(α, α) posterior mean,

$$\tilde{p} = \frac{w + \alpha}{G + 2\alpha} \in (0, 1), \tag{4}$$

and *assign Eq.* (3) *by outcome class* (wins $\to \widehat{A}_{\text{win}}$, losses $\to \widehat{A}_{\text{loss}}$, draws $\to 0$). This (i) keeps $\tilde{p}$ strictly inside $(0, 1)$, so there is no ε-corner; (ii) gives *every win in a group the same advantage and every loss the same advantage*, so any within-class tiebreak (the stall signal) is gone by construction; and (iii) is a **Rao–Blackwellisation** — the outcome class is the sufficient statistic for a binary reward, and conditioning on it replaces the noisy per-sample value with its exact conditional advantage, which is variance-optimal among unbiased estimators that respect the outcome. In 4p the outcome is graded (placement-linear), Eq. (3) no longer holds, and the code falls back to [B].

3 [B] Drop the per-group standard deviation (and use leave-one-out)

Variable reference — this section	
b_g	a per-group baseline subtracted from R_i
$\mu_{g,-i}$	leave-one-out group mean, Eq. (6)
$\mathcal{E}_g$	per-group gradient *energy* $\frac{1}{G}\sum_{i\in g}(\text{advantage})^2$
$p(1-p)$	the PFSP-"even" matchmaking weight (`rl_math_v9.tex`, league section)

For a constant-per-rollout advantage the variance-minimising baseline is the expected return, $b_g = \mathbb{E}[R \mid w_g] \approx \mu_g$; subtracting it is correct. **Dividing by σ_g is not part of any unbiased estimator** — it is a per-group reweighting, and Eq. (3) showed exactly what it does: it forces every group to unit energy. Compare the per-group energy with and without it:

$$\underbrace{\mathcal{E}_g^{/\sigma} = 1}_{\text{current (DeepSeek)}} \qquad \text{for all } \hat{p}; \qquad \underbrace{\mathcal{E}_g^{\text{center}} = \sigma_g^2 = 4B_o^2\,\hat{p}(1-\hat{p})}_{\text{center-only (Dr. GRPO)}}. \tag{5}$$

The center-only energy is **peaked at $\hat{p} = \frac{1}{2}$ and vanishes at the extremes** — it down-weights blowout/hopeless groups (nothing to learn) and concentrates the gradient on contested ones. This is precisely the right curriculum, and it is *the same function* $p(1-p)$ that PFSP-"even" uses to choose opponents. The league is already sampling matchups to maximise $p(1-p)$; the $/\sigma_g$ term flattens that back out, so matchmaker and update were optimising against each other. Removing it aligns them.

Implementation. `GRPO_STD_NORM=False` skips the division and applies *one global scale* to the batch (a single std over all transitions), which — unlike the per-group σ_g — preserves the cross-group $p(1-p)$ weighting while keeping the overall step size $O(1)$.

Leave-one-out baseline (RLOO). `GRPO_LOO=True` replaces μ_g by the mean of the *other* $G-1$ rollouts,

$$\mu_{g,-i} = \frac{1}{G-1}\sum_{j\in g,\, j\neq i} R_j \quad\Longrightarrow\quad R_i - \mu_{g,-i} = \frac{G}{G-1}\,(R_i - \mu_g). \tag{6}$$

The baseline no longer contains sample i, removing the $O(1/G)$ self-correlation that biases the included-mean estimator; the identity on the right shows the only effect on the *magnitude* is a benign $G/(G-1)$ rescale (here 16/15), so the value of LOO is the de-biasing, not a scale change.

4 [C] The shaping potential is a value, not a reward

Variable reference — this section	
$\Phi(s)$	shaping potential, $\Phi = \texttt{SHAPE\_SHIP}\,\varphi_{\text{ship}} + \texttt{SHAPE\_PROD}\,\varphi_{\text{prod}}$
$\varphi_{\text{ship}}, \varphi_{\text{prod}}$	log ship-margin / production-share vs. the strongest opponent (Ng potentials)
γ, λ	discount (`GAMMA`) and GAE trace (`GAE_LAMBDA`)
$V_t, \widehat{V}_t$	a value at s_t; the hand-built estimate Eq. (8)
a	fitted slope of outcome on terminal position change, Eq. (9)
$\delta_t, \widehat{A}_t$	TD residual and the per-step (GAE) advantage
L	rollout length; $\Delta\Phi_t = \Phi(s_t) - \Phi(s_0)$

GRPO's deepest weakness is that one trajectory-level scalar is smeared over all $T = 500$ steps: no *per-step* credit. We already compute a hand-crafted "who is winning" signal $\Phi(s)$ and feed it as potential-based reward shaping (PBRS). Under the Monte-Carlo return GRPO actually uses, that is nearly wasted. Telescoping the shaping increment $\gamma\Phi_{t+1} - \Phi_t$ over a rollout,

$$\sum_{t=0}^{L-1} \big(\gamma\Phi_{t+1} - \Phi_t\big) = \underbrace{-\Phi_0}_{\text{shared in group}} + (\gamma-1)\sum_{t=1}^{L-1}\Phi_t \;+\; \gamma\Phi_L. \tag{7}$$

The leading term $-\Phi_0$ is the world's *initial* imbalance — identical for all G members of a group (they share w_g, hence s_0) — so the group mean μ_g **cancels it exactly**. What survives is a tiny $(\gamma - 1) = -0.003$-weighted occupancy integral, not the credit signal we wanted. This is the textbook fact (Ng et al.; Wiewiora 2003) that *PBRS is equivalent to initialising the value function to Φ, and only does work in the presence of bootstrapping.* With no critic there is no bootstrapping, so the shaping is inert.

The fix: bootstrap with Φ. Use Φ as the value and run GAE. Anchor the value at the group baseline and let position change move it:

$$\widehat{V}_t^{\,i} = \mu_{g(i)} + a\left(\Phi(s_t^i) - \Phi(s_0^i)\right) = \mu_g + a\,\Delta\Phi_t^i, \tag{8}$$

with the slope fit once per iteration by closed-form least squares of the group-centred outcome on the *terminal* position change $\Delta\Phi_L$ (clamped to $[0, \texttt{GRPO\_PHI\_A\_MAX}]$):

$$a = \frac{\sum_i (R_i - \mu_g)\,\Delta\Phi_L^i}{\sum_i (\Delta\Phi_L^i)^2}. \tag{9}$$

Then standard GAE on the (now sparse: terminal-outcome-only) reward r_t,

$$\delta_t = r_t + \gamma\widehat{V}_{t+1} - \widehat{V}_t, \qquad \widehat{A}_t = \sum_{l\geq 0} (\gamma\lambda)^l\,\delta_{t+l}. \tag{10}$$

Two things fall out. At the root $\widehat{V}_0 = \mu_g$, so δ_0 subtracts the group baseline: **world difficulty is cancelled by the shared root**, recovering GRPO's whole purpose. And for an interior step ($r_t = 0$, $\gamma \approx 1$),

$$\delta_t \approx a\big(\Phi(s_{t+1}) - \Phi(s_t)\big), \tag{11}$$

so the per-step advantage credits exactly the moves that *improve your position*, with the terminal outcome propagated back through them by the $(\gamma\lambda)$ trace. Because $\widehat{V}$ is a function of state only, it is an *unbiased baseline* regardless of fit quality (a bad fit costs variance, never bias); the $\lambda < 1$ trace is the usual bias–variance knob. To avoid double-counting Φ, the explicit reward shaping is turned off in this mode (`GRPO_PHI_VALUE` routes Φ to the value, and the dense reward is set to zero).

5 [D] Common random numbers across a world-group

Variable reference — this section	
ξ_i, ζ_i	the *ego*'s and the *opponent*'s action-sampling noise in rollout i
w_g	the (shared) world; comets are deterministic per world
u, g_u	a uniform / Gumbel sample driving the categorical and the gate
Cov	covariance; the CRN variance identity below

A rollout's return is a function of three independent noise sources, $R_i = f(w_g, \xi_i, \zeta_i)$: the world (shared within a group), the ego's sampling ξ_i, and the neural opponent's sampling ζ_i. The group baseline is meant to isolate the effect of varying the *ego*'s actions, but with independent ζ_i it also absorbs opponent-sampling noise, inflating σ_g and loosening the advantage. **Common random numbers** couple the opponent across the group: draw one noise stream ζ_g and share it among all G members (the ego keeps independent ξ_i — that variation is the signal). By the control-variate identity

$$\mathrm{Var}(R_i - R_j) = \mathrm{Var}\, R_i + \mathrm{Var}\, R_j - 2\ \mathrm{Cov}(R_i, R_j), \tag{12}$$

raising Cov between groupmates (via shared ζ) lowers the variance of within-group differences, hence of $\widehat{A}_i$. Concretely the categorical **WHERE** is sampled by Gumbel-max with noise shared across the group, and the **GATE** by a shared uniform:

$$d_s^i = \arg\max_d \big(\ell_{s,d}^i + g_u[g(i), s, d]\big), \qquad f_s^i = \mathbb{1}\big[u[g(i), s] < p_s^i\big], \tag{13}$$

where g_u, u are indexed by the *group* $g(i)$, not the rollout — identical within a group, independent across groups. Identical-logit groupmates therefore sample identical actions (a clean unit test); when logits differ (the ego has diverged the state) the actions differ but the *noise* is common, which is the variance reduction. CRN applies only to neural opponents (scripted seats are handled in the engine); the world and its comets are already shared by construction.

6 [C1] The rank advantage: bounded and reward-scale-free

Verdict: this FAILED *— the rank advantage collapsed the agent into passivity in the live ablation (Section 14). The derivation below is the mechanism, not a recommendation.*

Variable reference — this section	
r_i	within-group rank of R_i (average rank on ties), $r_i \in [0, G-1]$
c_G	std of a uniform rank, $c_G = \sqrt{(G^2-1)/12}$

Section 2 showed the std-normalised win advantage is $\widehat{A}_{\text{win}} = \sqrt{(1-\hat{p})/\hat{p}}$, which **diverges** as $\hat{p} \to 0$. That is the precise engine of the second collapse: once the snapshot pool is strong the learner sits in near-hopeless groups (small $\hat{p}$), and a single lucky win is handed an arbitrarily large advantage, so the policy chases it and abandons the launches that lose "honestly" — the passivity ratchet. [A]'s shrinkage caps the divergence at a finite $\sqrt{(1-\tilde{p})/\tilde{p}}$, but the worst case still grows with how lopsided the matchup is.

The **rank advantage** drops the dependence on $\hat{p}$ altogether: replace R_i by its within-group rank and standardise to unit variance,

$$\widehat{A}_i = \frac{r_i - \frac{G-1}{2}}{c_G}, \qquad c_G = \sqrt{\frac{G^2-1}{12}} \tag{14}$$

(average rank for ties; $O(G^2)$ by pairwise comparison — $G = 16$, fully vectorised, no host sync). Two properties:

- **Bounded for every group.** The extreme rank gives $|\widehat{A}_i| \le \dfrac{(G-1)/2}{c_G} = \sqrt{\dfrac{3(G-1)}{G+1}} \xrightarrow{G\to\infty} \sqrt{3}$ ($= 1.63$ at $G = 16$), *independent of* $\hat{p}$. The lone win in a hopeless group gets $+1.63$, not $+\infty$.

- **Invariant to the reward scale.** Eq. (14) depends only on the *ordering* of $\{R_i\}$, so it is unchanged by any monotone transform of the return — `PPO_REWARD_SCALE`, the `WIN_DECAY/LOSS_DECAY` terms, every `SHAPE_*/CAPTURE_*` magnitude. The heavily hand-tuned reward constants stop steering the gradient.

This is the rank-shaping of natural evolution strategies (Wierstra et al. 2014): a small population's raw fitness scale is unreliable, only its ordering is — and the ordinal objective is what 4p placement scores anyway. **Trade-off vs. [B]:** rank gives every group unit energy ($\mathcal{E}_g = 1$, like the $/\sigma_g$ path of Eq. (5)), so it does *not* restore the cross-group $p(1-p)$ weighting that center-only [B] does. Rank fixes the *within-group* unboundedness; [B] fixes the *cross-group* difficulty inversion — different axes. In code `GRPO_RANK_ADV` short-circuits the center-only path, so the trajectory estimator is rank *or* Dr. GRPO+RMS (the two anti-collapse bundles `ANTICOLLAPSE` / `DRGRPO_BUNDLE` in `grpo_diag`).

7 Tier 0: the trust region and the advantage scale

Three cheap levers on the *update* (not the advantage estimator) that compose with everything.

[D3] Clip-higher (`CLIP_HI`). The symmetric clip $\mathrm{clip}(\rho, 1-\varepsilon, 1+\varepsilon)$ caps how fast a currently-unlikely action's probability can rise; with `INIT_GATE_BIAS`$= -1$ (few planets fire) that throttles exactly the launch exploration we need. DAPO's clip-higher relaxes only the upper bound, $\mathrm{clip}(\rho,\, 1-\varepsilon,\, 1+\varepsilon_{\mathrm{hi}})$ with $\varepsilon_{\mathrm{hi}} =$ `CLIP_HI` $> \varepsilon$, so positive-advantage probability-raising passes while the downside stays tight. `CLIP_HI`$= 0$ recovers the symmetric clip.

[D1] Length-normalised ratio (`RATIO_LENGTHNORM`). The per-step log-ratio is a *sum* over owned planets, $\ell = \sum_{i\in\mathrm{owned}} \Delta \log p_i$, so $\mathrm{Var}\,\ell \propto n_{\mathrm{owned}}$ and the fixed `CLIP/KL_TARGET` bite hardest in many-planet (midgame) states. GSPO clips the geometric-mean ratio $s = (\pi_\theta/\pi_{\theta_{\mathrm{old}}})^{1/n_{\mathrm{owned}}} = \exp\big(\frac{1}{n_{\mathrm{owned}}} \sum_i \Delta \log p_i\big)$, making the trust region a function of policy change *per decision*, not board size (it also realigns the surrogate with the entropy term, already a per-owned-source mean). The KL early-stop reuses the same normalised log-ratio.

[D4] Pinned advantage scale (`ADV_TARGET_RMS`). On the center-only path ([B], `GRPO_STD_NORM=False`) the single global scale is a *batch* std that drifts with league difficulty iteration-to-iteration. Since `ADAM_EPS`$= 10^{-5}$ is not negligible against these small sparse-reward gradients, that drift acts as a stealth learning-rate schedule keyed to which opponent was drawn. `ADV_TARGET_RMS`> 0 rescales the centred advantage to a *constant* RMS, $\widehat{A} \leftarrow (\widehat{A} - \overline{\widehat{A}}) \cdot$ `ADV_TARGET_RMS`$/\,\mathrm{rms}(\widehat{A})$, so the step size means the same thing every iteration (one global scalar; still preserves the cross-group $p(1-p)$ weighting).

8 [A1] Rao–Blackwellise the launch gate

Variable reference — this section	
f_i, p_i	realised fire bit $\in \{0,1\}$ and its probability for owned planet i
ℓ_i^{w}	WHERE log-prob $\log p_{\mathrm{where},i}(d_i)$ of the sampled destination

The factored log-prob is $\log \pi_\theta = \sum_i [\log p_{\mathrm{gate},i}(f_i) + f_i\, \ell_i^{\mathrm{w}}]$: the destination score is gated by the *sampled* bit f_i, so for the $1-p_i$ fraction of planets that did not fire the WHERE gradient is identically zero this sample — pure Monte-Carlo noise on the strategically decisive head. Conditioning on the gate (Rao–Blackwell) replaces f_i by its expectation, $\mathbb{E}_{f_i \sim \mathrm{Bern}(p_i)}[f_i \nabla \ell_i^{\mathrm{w}}] = p_i \nabla \ell_i^{\mathrm{w}}$, removing the gate's sampling variance $p_i(1-p_i)\|\nabla \ell_i^{\mathrm{w}}\|^2$ *exactly* (the gate is a known Bernoulli — no estimation). Implemented as a straight-through so the importance ratio stays honest: the *forward* value is the realised $f_i \ell_i^{\mathrm{w}}$ (the clip sees the true ratio) while the *gradient* flows as the de-noised $p_i \nabla \ell_i^{\mathrm{w}}$,

$$\widetilde{\ell_i^{\mathrm{w}}} = \big(f_i\, \ell_i^{\mathrm{w}}\big)^{\downarrow} + \bar{p}_i\, \ell_i^{\mathrm{w}} - \big(\bar{p}_i\, \ell_i^{\mathrm{w}}\big)^{\downarrow}, \qquad (\cdot)^{\downarrow} = \text{stop-grad}, \quad \bar{p}_i = p_i^{\downarrow}. \tag{15}$$

This is the discrete-sub-action marginalisation of expected policy gradients (Ciosek–Whiteson; "mean actor-critic"). The Tucker et al. (2018) "mirage" caveat is about *learned* action-dependent baselines, not the exact analytic marginalisation of a known Bernoulli, which is strictly variance-reducing at zero extra forward cost. The gate keeps its own realised score (no double count).

9 [C2] A risk-sensitive (CVaR) baseline

The flat advantage optimises the group *mean*, but the ladder is decided by not getting exploited by the *strong* opponents — a mean-vs-tail mismatch (the recurring "great training Elo, loses head-to-head"). Optimise the worst-α tail: with the empirical α-quantile q_α and $\mathrm{CVaR}_\alpha = \frac{1}{\alpha G} \sum_{R_i \le q_\alpha} R_i$, baseline on the CVaR and up-weight the tail,

$$\widehat{A}_i = \big(R_i - \mathrm{CVaR}_\alpha\big)\,\big(1 + \lambda \mathbb{1}[R_i \le q_\alpha]\big), \tag{16}$$

(α = `GRPO_CVAR_ALPHA`, λ = `GRPO_CVAR_LAMBDA`; `kthvalue` on the G axis, one fused op). This is the CVaR policy gradient (Tamar; Chow); it pairs with [D] CRN, which makes the optimised tail a pure function of *ego* variance. **Risk:** over-pessimism re-summons the passivity basin — start mild ($\alpha \ge 0.5$, $\lambda \approx 1$) and watch `lnch/st` against the tripwire.

10 [B1] Per-step credit from shared-root siblings

[C] manufactures per-step credit from a hand-built Φ with one global OLS slope. The G rollouts of a group are a richer, free source: they share s_0 and branch only on the ego's actions — a tree rooted at the world (the VinePPO setting, but the branches are already paid for). Bucket the siblings by a quantised position signal at each step ($q_t = \lfloor \varphi_{\mathrm{ship}}(s_t)/\Delta \rceil$, Δ = `GRPO_SIBLING_BUCKET_DELTA`) and use the matched-bucket mean *terminal* return as a per-step value, falling back to the whole-group mean when a bucket is thinner than `GRPO_SIBLING_MIN_PEERS`:

$$\widehat{V}(s_t^i) = \operatorname*{mean}_{j:\, q_t(j) = q_t(i)} R_j, \qquad \widehat{A}_t = \mathrm{GAE}_{\gamma,\lambda}\big(r_t, \widehat{V}\big). \tag{17}$$

Early steps (all siblings in one bucket) reduce to flat GRPO — correct at the root; late steps separate the siblings that pulled ahead from those that fell behind *at matched position*, the per-step signal the flat broadcast destroys. No critic, no extra rollout; composes with CRN (matched siblings then also share opponent noise).

11 [B2] The dead value head as a representation aux

Under GRPO the policy net's value head is never trained. `GRPO_AUX_VALUE` trains it by MC regression to the rollout return (PopArt-normalised), $\mathcal{L}_{\mathrm{aux}}$ = `GRPO_AUX_COEF` $\cdot\, (V_\theta(s_t) - \mathrm{PopArt}(R_i))^2$, **as an auxiliary only** — it never enters the advantage, so the update stays critic-free and the weights stay interchangeable with a PPO checkpoint. The signal ("which states lead to winning") regularises the shared trunk toward outcome-relevant features; keep the coefficient small so it does not fight the policy. (This is the safe cousin of the concurrent UNREAL `AUX_REWARD_PRED` head, which regresses the immediate reward on the PPO path.)

12 [E1]/[E2] Opponent-conditioned baseline (one wired, one reserved)

Each iteration fills the whole batch with *one* PFSP-sampled opponent, so a group's mean return depends on which foe was drawn — a between-iteration variance/bias source the world-baseline cannot see. The fix is a control variate on opponent strength: subtract the league's own Elo-expected outcome before centering,

$$R_i \leftarrow R_i - \texttt{GRPO\_OPP\_BASELINE\_W} \cdot (2\hat{E}_o - 1)\, B_o, \qquad \hat{E}_o = \left(1 + 10^{(e_o - e_{\text{ego}})/s}\right)^{-1}, \tag{18}$$

so the advantage reads "better than the rating system predicted vs. *this* foe." **Caveat (important).** With one opponent per iteration $\hat{E}_o$ is *constant within every group*, so per-group centering already removes it: **[E1] is a no-op on its own**. It only bites once groups *span* opponents — which is **[E2]** `GRPO_OPP_STRATA`: sample K opponents per iteration, one per contiguous block of groups, so the baseline averages the strength axis. [E2] needs a per-block opponent rollout (each block evaluates a different opponent net), an invasive change to the rollout's per-step opponent loop; it is **reserved, not wired** (Section 15). [E1]'s plumbing is in place as its companion.

13 The proof: a deterministic advantage-bound probe

Caveat: this proves the advantage is BOUNDED, not that bounded helps. *The live ablation (Section 14) shows the bounded rank advantage* causes *passivity — boundedness looked necessary but is not sufficient, and was in fact the wrong objective. Keep this section as "what the estimator does," not "why to use it."* The anti-collapse claim is checked on CPU, without training, by `grpo_diag.advantage_bound_probe`, which runs the *same* `grpo_flat_advantage` the trainer uses over synthetic groups of every win-count $k = 0..G$ and reports the worst $|\widehat{A}|$:

estimator	worst $\lvert\widehat{A}\rvert$ ($G = 16$)	behaviour vs. lopsidedness
`legacy_std`	3.75	grows $\sim \sqrt{(1-\hat{p})/\hat{p}}$ as $\hat{p} \to \{0, 1\}$
`analytic[A]`	2.83	shrunk, but still grows
`rank[C1]`	**1.63**	**flat** $= \sqrt{3(G-1)/(G+1)}$, for every group

A companion difficulty-inversion table (groups of mixed win-rate in one batch) shows `legacy_std` forcing equal per-group energy while center-only+RMS keeps the contested groups weighted. `run_anticollapse_co` then runs a short three-arm A/B (`legacy / rank[C1] / drgrpo[B]`) logging advantage-health, launch-rate, and return; passing `resume_from` a strong checkpoint reproduces the real spiral on a GPU. The whole thing is the generated notebook `setup1_v12_grpo_anticollapse.ipynb` (`python scripts/make_v12_nb.py --grpo-smoke`).

14 Empirical result: the local ablation contradicts the theory

The Tier 0–2 estimators above are NOT validated — in a controlled GPU ablation they made things *worse*. On an RTX 3070 Ti (`scripts/_gpu_ab*.py`: h256 trunk, BC-warm shared init, identical seed/worlds, 2p, 100 iters/arm) only **legacy per-group std-norm stayed stable**; every "anti-collapse" estimator drove the agent into the project's *passivity basin* (launch-rate $\to 0$, Elo down):

config	launch/st	Elo2	verdict
legacy std-norm (symmetric decay)	$0.70 \to 0.59$	$1497 \to$ **1576**	**stable / improving**
rank[C1] + clip-hi	$0.53 \to 0.10$	$\to 1259$	passivity
rank[C1] no-clip	$0.48 \to 0.27$	$\to 1292$	passivity
drgrpo[B] center-only	$0.57 \to 0.34$	$\to 1370$	passivity
drgrpo[B] + clip-hi	$0.46 \to 0.03$	$\to 1274$	worst
sibling[B1]	$0.57 \to 0.24$	$\to 1295$	passivity ($\lvert\widehat{A}\rvert_{\max} = 16$)
phi-value[C]	$0.59 \to 0.31$	$\to 1358$	passivity ($\lvert\widehat{A}\rvert_{\max} = 22$)
legacy + "lose-slower" decay (`LOSS_DECAY` < 1)	$0.55 \to 0.20$	$\to 1305$	**COLLAPSED**

Two conclusions, both confirmed:

1. **Bounded $\neq$ good.** The $\sqrt{3}$ bound (Section 6) held *exactly*, but the same property that tames the lucky-win chase — discarding win-*magnitude* — also removes the signal that rewards *decisive aggression*. The std rare-win amplification of Eq. (3) is double-edged: it is the it426 spiral amplifier *and* the thing that keeps the policy attacking. rank/drgrpo/sibling/phi-value all weaken it and collapse into passivity; only std-norm keeps both.
2. **The it426 collapse was the reward TRIGGER, not the estimator.** The *identical* legacy std-norm is stable under symmetric decay and collapses under an asymmetric "lose-slower" decay (last row). So Section 1's "amplifier" framing was half the story: removing the amplifier (Sections 2, 3, 6) is the *wrong* fix; **removing the trigger** (keep `WIN_DECAY` = `LOSS_DECAY` = 1) is the right one, and the v12 default already does.

Recommendation: train on legacy std-norm with symmetric decay (the battle-tested v9/v12 default; the `setup1_v12_grpo_a100.ipynb` preset is reverted to it). The estimator flags stay in the code for research but must be re-ablated on the 3070 Ti before any A100 use; `train.py` now prints the active estimator and warns on the mutually-exclusive-flag footgun (the "all flags on" run that silently selected sibling/phi-value). If the spiral ever recurs under legacy, the lever is **trust-region/entropy** (KL-to-reference `GRPO_KL_COEF` > 0; an entropy floor), not an estimator swap.

15 Pending: Tier 3 structural changes

Designed but *not implemented*; each is a larger surface than a flag flip. Listed by expected value.

[E2] Opponent stratification. The per-block multi-opponent rollout that makes [E1] active: sample K opponents, run each on its block of groups, update Elo per stratum. Cuts the opponent-draw variance $\sim 1/K$ and gives $K\times$ cleaner ratings. Touches the rollout opponent loop and the Elo update.

[E3] Seat-antithetic mirror pairing. Within a group, pair rollouts that *swap* which seat the ego plays (the deterministic seat-swap tiling already exists). In a near-zero-sum game $O(\text{ego}=0) \approx -O(\text{ego}=1)$, so the pair-sum cancels world *and* absolute-seat advantage — an antithetic control variate that also kills the documented seat-0 positional overfit. 2p first (the 4p swap is a 4-cycle, not an involution).

[A2] Antithetic ego coupling. Extend CRN [D] to the ego: antithetic gate Bernoullis (U and $1 - U$) across paired groupmates give negatively-correlated returns, lowering the variance of the G-sample baseline for free (throughput, not compute, is the binding constraint).

[D2] Per-factor clipping. Clip each planet's $\Delta \log p_i$ then sum, instead of clipping the joint product ratio — the token-level end of the GSPO/PPO spectrum at the correct (per-planet) granularity, so one bad planet can neither escape nor veto the clip. Needs a per-source log-prob from the distributions.

[D5] Adaptive KL-to-reference. Replace the bang-bang KL early-stop with a smooth penalty whose β is driven by a PI controller to a KL setpoint (the $k3$ machinery is already present); a better-conditioned trust region that makes more update epochs safe.

[C3]/[E4] Calibrated / Elo-graded outcome. Replace $\text{sign}(\cdot)$ by a saturating tanh ship-margin (training-time densification of the one informative scalar) and/or grade the outcome by the Elo residual so beating a strong foe counts more. Saturation + decay guard against score-running; report only the binary outcome.

[B3] Per-factor counterfactual credit (COMA). Treat each owned planet as an agent; give its gate/WHERE a counterfactual advantage from sibling launch-event differencing ("which launch won it"). Builds on [B1]'s sibling buckets.

[B4] Hindsight credit assignment. Return-conditioned HCA reweighting $(1 - \pi_\theta/h)\, R$ with a small hindsight head $h(a \mid s, R)$; the factored action makes h cheap. The principled version of [B1].

[E5] Worst-case / PSRO weighting. Weight each iteration's advantage by $(1 - \hat{E}_o)^\beta$ (focus on foes you cannot yet beat) $\rightarrow$ a min-exploitability objective, the principled answer to the non-transitive cycling that average-league-return induces. The full version solves a Nash meta-distribution over the league payoff matrix (PSRO).

Every Tier 3 item must be judged on **held-out head-to-head**, never training Elo — the project's standing lesson (a high-Elo agent that finishes last in H2H is the failure these all target).

16 Flags, defaults, and interactions

flag	default	effect when changed
`GRPO_STD_NORM`	`True`	`False` $\to$ center-only + global scale (Dr. GRPO)
`GRPO_LOO`	`False`	`True` $\to$ leave-one-out group mean (RLOO)
`GRPO_ANALYTIC_ADV`	`False`	`True` $\to$ closed-form 2p advantage at shrunk $\tilde{p}$ (4p falls back)
`GRPO_ANALYTIC_ALPHA`	`1.0`	Beta pseudocount in Eq. (4)
`GRPO_PHI_VALUE`	`False`	`True` $\to$ Φ-as-value GAE (per-step credit); shaping reward off
`GRPO_PHI_A_MAX`	`3.0`	clamp on the fitted slope a, Eq. (9)
`GRPO_CRN`	`False`	`True` $\to$ couple neural-opponent sampling within a group
Tier 0–2 anti-collapse (Part II)		
`CLIP_HI`	`0.0`	$> 0 \to$ asymmetric upper clip $1 +$ `CLIP_HI` (DAPO) [D3]
`RATIO_LENGTHNORM`	`False`	`True`$\to$ divide the log-ratio by #owned planets (GSPO) [D1]
`ADV_TARGET_RMS`	`0.0`	$> 0 \to$ pin the centred-advantage RMS to this constant [D4]
`GRPO_RANK_ADV`	`False`	`True`$\to$ within-group rank advantage, bounded $\sqrt{3}$ [C1]
`GRPO_RB_GATE`	`False`	`True`$\to$ Rao–Blackwellise the gate (de-noise WHERE grad) [A1]
`GRPO_CVAR_ALPHA`	`0.0`	$> 0 \to$ CVaR worst-α tail baseline [C2]
`GRPO_SIBLING_BASELINE`	`False`	`True`$\to$ per-step shared-root sibling GAE [B1]
`GRPO_AUX_VALUE`	`False`	`True`$\to$ train the value head as a representation aux [B2]
`GRPO_OPP_BASELINE_W`	`0.0`	$> 0 \to$ Elo-expected opponent baseline (no-op w/o [E2]) [E1]
`GRPO_OPP_STRATA`	`1`	*reserved* — per-block multi-opponent rollout [E2]

Interactions. [A] and [B] both shape the *flat* trajectory advantage and are mutually exclusive for the terminal term: with [A] on (2p) the analytic values are used and the std/LOO knobs of [B] only toggle the optional global whiten. **[C1] rank also takes the flat path and short-circuits [A]/[B]** — it is the estimator, so pick *one* of {rank, analytic, std/center}. [C] *replaces* the flat path with the GAE path, so [A]/[C1]'s values do not apply while [C] is on (the group baseline enters [C] through μ_g in Eq. (8), and `GRPO_LOO` still selects $\mu_{g,-i}$ there); **[B1] sibling** likewise replaces the flat path with its own per-step GAE. **Tier 0** ([D3]/[D1]/[D4]) acts on the update/scale and composes with every estimator; **[A1]** de-noises the WHERE gradient under any estimator; **[B2]** only adds an aux loss; **[D]** changes how the *rollout* is sampled — all orthogonal. Two coherent anti-collapse bundles (`grpo_diag`): `ANTICOLLAPSE`={[C1] rank, [D3]} (bounded, scale-free) and `DRGRPO_BUNDLE`={[B] center-only+LOO, [D4], [D3]} (cross-group weighting). Recommended first run: `ANTICOLLAPSE` (highest-confidence; judge on held-out H2H, not training Elo), then compose [A1]+[D1] for variance, or move to [B1]/[C] for per-step credit.